Global Coding Bootcamp Platform User Documentation

Module: ITMDA0-11 | Database Management (MySQL) | Project 1 | 2026

1. Overview

The Global Coding Bootcamp Platform is a MySQL relational database system that manages learner accounts, mentors, bootcamp courses, enrollments, learning progress, assessments, digital resources, and system error logging.

2. Database Setup Instructions

Run the SQL script files in the following order using MySQL Workbench:

Step	File	Description
1	01_create_database.sql	Creates the database and all tables
2	02_insert_data.sql	Inserts sample data
3	03_views.sql	Creates all views
4	04_stored_procedures.sql	Creates all stored procedures
5	05_triggers.sql	Creates all triggers
6	06_indexes.sql	Creates all indexes

3. Database Schema Summary

Table	Primary Key	Description
learners	learner_id	Learner accounts and subscription info
mentors	mentor_id	Instructor profiles
bootcamps	bootcamp_id	Course/bootcamp definitions
enrollments	enrollment_id	Learner-to-bootcamp enrolment records
modules	module_id	Learning modules within bootcamps
module_progress	progress_id	Per-learner module completion tracking
assessments	assessment_id	Quizzes, projects, coding challenges
assessment_results	result_id	Learner marks and skill scores
resources	resource_id	Digital learning resources
learner_resources	lr_id	Resources downloaded by learners
error_log	log_id	System error and event audit log

4. Views

vwBlockedLearners

Shows all learners whose subscriptions have expired and whose accounts are blocked.

```
SELECT * FROM vwBlockedLearners;
```

vwTopCoders

Returns the top 20 learners ranked by total skill score accumulated across all assessments.

```
SELECT * FROM vwTopCoders;
```

vwMostDownloadedResources

Shows the top 20 most downloaded resources, ordered by download count.

```
SELECT * FROM vwMostDownloadedResources;
```

vwPopularBootcamps

Shows the top 5 bootcamps by enrollment count, including capacity usage percentage.

```
SELECT * FROM vwPopularBootcamps;
```

5. Stored Procedures

spRegisterLearner

Registers a new learner. Validates that both the username and email are unique before inserting.

Parameter	Type	Description
p_username	VARCHAR(50)	Unique username
p_password	VARCHAR(255)	Password (auto-hashed with SHA2-256)
p_email	VARCHAR(100)	Learner email address
p_expiry_date	DATE	Initial subscription expiry date
p_storage_limit	DECIMAL(10,2)	Storage allocation in MB

```
CALL spRegisterLearner('newuser', 'SecurePass1!', 'new@mail.com', '2027-01-01', 50
0.00);
```

spRenewSubscription

Adds days to a learner's subscription and automatically reactivates blocked accounts.

Parameter	Type	Description
p_learner_id	INT	ID of the learner to renew
p_days_to_add	INT	Number of days to extend the subscription

```
CALL spRenewSubscription(3, 365);
```

spEnrollLearner

Enrolls a learner in a bootcamp. Checks account status, available capacity, and duplicate enrollments.

Parameter	Type	Description
p_learner_id	INT	ID of the learner
p_bootcamp_id	INT	ID of the bootcamp to enroll in

```
CALL spEnrollLearner(1, 3);
```

spAddResource

Records a resource download for a learner. Validates that the learner's storage limit will not be exceeded.

Parameter	Type	Description
p_learner_id	INT	ID of the learner
p_resource_id	INT	ID of the resource to download

```
CALL spAddResource(2, 5);
```

spSendNotice

Generates a subscription status notice showing remaining days and account status for a learner.

Parameter	Type	Description
p_learner_id	INT	ID of the learner

```
CALL spSendNotice(1);
```

6. Triggers

Trigger	Event	Purpose
trg_block_expired_learner_insert	BEFORE INSERT on learners	Blocks new accounts inserted with a past expiry date
trg_block_expired_learner_update	BEFORE UPDATE on learners	Blocks accounts when subscription_expiry is updated to a past dat
trg_log_learner_blocked	AFTER UPDATE on learners	Logs to error_log when an account status changes to Blocked
trg_log_enrollment_insert	AFTER INSERT on enrollments	Logs a capacity warning when a bootcamp reaches 80% or more f
trg_log_storage_warning	AFTER INSERT on learner_resources	Logs a storage warning when a learner uses 90% or more of their
trg_log_assessment_submission	AFTER INSERT on assessment_results	Logs an alert when a learner scores below 50% on an assessment

7. Indexes

Table	Indexed Column(s)	Purpose
learners	username	Fast login and username lookup
learners	email	Unique check on registration
learners	account_status, subscription_expiry	Fast blocked learner queries

Table	Indexed Column(s)	Purpose
bootcamps	category, level	Filtering bootcamps by type and level
bootcamps	mentor_id	JOIN performance with mentors table
enrollments	learner_id, bootcamp_id	Fast enrollment lookups in both directions
assessment_results	learner_id, skill_score	vwTopCoders aggregation performance
resources	download_count, resource_type	vwMostDownloadedResources ranking
error_log	learner_id, error_category, logged_at	Audit log queries and reporting

8. Error Handling

All stored procedures validate required inputs and raise informative SIGNAL errors using SQLSTATE '45000'. Failed operations are also written to the error_log table automatically for auditing purposes.

To view recent errors:

```
SELECT * FROM error_log ORDER BY logged_at DESC LIMIT 20;
```

9. Backup and Restore

9.1 Creating a Backup

Use MySQL Workbench's built-in Data Export tool as follows:

- Open MySQL Workbench and connect to your server.
- Click **Server** in the top menu and select **Data Export**.
- Tick **global_bootcamp** in the schema list on the left.
- Select **Export to Self-Contained File** and click **Browse** to choose a save location.
- Ensure the following options are all ticked at the bottom of the screen:
  - Include Create Schema
  - Dump Stored Procedures and Functions
  - Dump Events
  - Dump Triggers
- Click **Start Export**.

Note: It is essential to tick "Dump Stored Procedures and Functions" and "Dump Triggers". Without these options, stored procedures and triggers will not be included in the backup file.

9.2 Restoring and Testing a Backup

To verify the backup restores correctly:

- Click **Server** and select **Data Import**.
- Select **Import from Self-Contained File** and browse to the backup file.
- Click **Start Import**.
- Verify the restore was successful by running the following queries:

```
USE global_bootcamp;
SHOW TABLES;
SHOW PROCEDURE STATUS WHERE db = 'global_bootcamp';
```

- You should see all 11 tables and all 5 stored procedures listed.

The backup file (global_bootcamp_backup.sql) is included in the submission zip.

10. ER Diagram

The ER diagram (ITMDA0-11_ER_Diagram.pdf) is included in the submission and illustrates all entities, attributes, primary keys, foreign keys, and relationships between tables.

Key relationships:

- A **mentor** teaches many **bootcamps** (1:N)
- A **bootcamp** has many **modules** and **assessments** (1:N)
- A **learner** enrolls in many **bootcamps** via **enrollments** (M:N)
- A **learner** tracks progress in many **modules** via **module_progress** (M:N)
- A **learner** submits many **assessment_results** (M:N)
- A **learner** downloads many **resources** via **learner_resources** (M:N)
- **error_log** optionally links to a **learner** (nullable FK, 0:N)